Final Reflection

1) Cancer Genomics Dataset
* What features are present in the cancer genomics dataset?
* What preprocessing steps did you apply (if any)?
* What initial observations did you make about the dataset (dimensionality, class balance, missingness, feature scale, etc.)?

The features present in this dataset are genomic signals from different gene loci. I did not apply any preprocessing steps as there were no missing values in any of the rows. I noticed that there are 5479 features (different gene loci) and patient data from 801 people. Additionally, there are 5 classification categories: Breast Invasive Carcinoma, Kidney Renal Clear Cell Carcinoma, Lung Adenocarcinoma, Prostate Adenocarcinoma, and Colon Adenocarcinoma. The categories are imbalanced with most true labels being BRCA, KIRC, LUAD, and PRAD being around the same, and COAD being about half the samples of those three. Additionally all of the features' values are floats.

```
print(X_cancer.shape, y_cancer.value_counts().to_dict())
print(X_cancer.isna().sum())
```

✓ 0.5s

```
Successfully loaded heart disease data with 1025 rows
(801, 5479) {'BRCA': 300, 'KIRC': 146, 'LUAD': 141, 'PRAD': 136, 'COAD': 78}
gene_18       0
gene_21       0
gene_26       0
gene_28       0
gene_29       0
              ..
gene_20500    0
gene_20504    0
gene_20514    0
gene_20524    0
gene_20529    0
Length: 5479, dtype: int64
```

2) Model Performance (Heart Disease + Cancer Genomics)
For **both datasets**, report and discuss model performance using metrics like accuracy, precision, recall, F1-score.

Include figures/tables of cross-validation results for each dataset, and briefly compare:
* Which dataset was easier/harder to classify?
* Which errors were most common (especially in the cancer dataset)?

Heart Disease Classification	Mean	Std
Accuracy	0.922	0.020
Precision	0.924	0.021
Recall	0.923	0.020
F1	0.923	0.020
ROC_AUC	0.975	0.015

Cancer Type Classification	Mean	Std
Accuracy	0.985	0.011
Precision	0.989	0.007
Recall	0.981	0.011
F1	0.985	0.010
ROC_AUC	0.999	0.000

The Gradient Boosting Classifier did a much better job at predicting the cancer type dataset than the heart disease dataset. The average cross-validation accuracy, precision, recall, and f1 score of the cancer type classification model were all 6% higher than the heart disease classification model. The area under the ROC was

also 2% higher. The most common error in the Cancer dataset was misclassifying Prostate Adenocarcinoma as Breast Invasive Carcinoma.

3) Hyperparameter Tuning (Gradient Boosting)
Describe your hyperparameter tuning process and results:
* Which hyperparameters did you explore (e.g., `max_depth`, `n_estimators`, `learning_rate`, etc.)?
* What range of values did you test?
* Which configuration performed best on each dataset, and why do you think it worked well?
* Provide grid search table or plot.

```
param_grid = {
    'max_depth': [3, 10, 45],
    'n_estimators': [50, 100, 150],
    'learning_rate': [0.1, 0.001, 0.0001, 0.5]
}
```

For both datasets I ran GridSearch cv on these three hyperparameters (max_depth, n_estimators, and learning_rate). I tested max depth on the values 3, 10, and 45. I tested n_estimators on the values 50, 100, and 150. Finally, I tested Learning Rate on the values 0.1, 0.001, 0.0001, and 0.5. For the heart disease dataset the best parameter combination found was 0.5 step size for the learning rate, a max tree depth of 10, and 50 trees (estimators). For the cancer type dataset the best parameter combination was 0.1 step size for the learning_rate, 3 for the max tree depth, and 150 trees. In the heart disease dataset it seems like a greater depth led to the model learning more complex patterns in the data and needing less overall estimators to come up with good results. On the other hand the cancer dataset had an ideal depth of 3 and actually did worse with more indicating the model might have started to overfit with greater depths. With this decrease in depth the model needed significantly more trees (compared to the heart disease dataset) to achieve its best results.

Heart Disease Classification

Learning Rate	Max Depth	n_estimators	Mean Test Score	Rank
0.1	3	50	0.975087	30
0.1	3	100	0.982551	21
0.1	3	150	0.984991	11
0.1	10	50	0.988701	7
0.1	10	100	0.985365	9
0.1	10	150	0.984374	13
0.1	45	50	0.988701	7
0.1	45	100	0.985365	9
0.1	45	150	0.984374	13
0.001	3	50	0.896023	36
0.001	3	100	0.900886	32

0.001	3	150	0.903328	31
0.001	10	50	0.983858	16
0.001	10	100	0.984322	15
0.001	10	150	0.981692	26
0.001	45	50	0.983858	16
0.001	45	100	0.984579	12
0.001	45	150	0.981709	25
0.0001	3	50	0.896488	33
0.0001	3	100	0.896488	33
0.0001	3	150	0.896488	33
0.0001	10	50	0.981714	24
0.0001	10	100	0.981284	28
0.0001	10	150	0.982385	22

0.0001	45	50	0.981611	27
0.0001	45	100	0.981284	28
0.0001	45	150	0.982385	22
0.5	3	50	0.983688	18
0.5	3	100	0.983321	19
0.5	3	150	0.983254	20
0.5	10	50	0.991089	1
0.5	10	100	0.991089	1
0.5	10	150	0.991089	1
0.5	45	50	0.990245	4
0.5	45	100	0.990245	4
0.5	45	150	0.990245	4

Cancer Type Classification

Learning Rate	Max Depth	n_estimators	Mean Test Score	Rank
0.1	3	50	0.999855	2
0.1	3	100	0.999851	5
0.1	3	150	0.999884	1
0.1	10	50	0.999806	6
0.1	10	100	0.999806	8
0.1	10	150	0.999855	3
0.1	45	50	0.999806	6
0.1	45	100	0.999806	8
0.1	45	150	0.999855	3
0.001	3	50	0.999201	32
0.001	3	100	0.998975	33
0.001	3	150	0.998661	35

0.001	10	50	0.999292	30
0.001	10	100	0.999370	21
0.001	10	150	0.999684	20
0.001	45	50	0.999292	29
0.001	45	100	0.999370	21
0.001	45	150	0.999698	16
0.0001	3	50	0.998637	36
0.0001	3	100	0.998942	34
0.0001	3	150	0.999247	31
0.0001	10	50	0.999298	27
0.0001	10	100	0.999354	23
0.0001	10	150	0.999347	25
0.0001	45	50	0.999298	27

0.0001	45	100	0.999327	26
0.0001	45	150	0.999354	24
0.5	3	50	0.999687	17
0.5	3	100	0.999687	17
0.5	3	150	0.999687	17
0.5	10	50	0.999795	10
0.5	10	100	0.999795	10
0.5	10	150	0.999795	10
0.5	45	50	0.999795	10
0.5	45	100	0.999795	10
0.5	45	150	0.999795	10

4) Feature Importance and Interpretation
Discuss which features were most predictive:

* For gradient boosting: report the top features by importance (and include a figures/table/plot).
* Compare feature importance to HW1 feature selection (for example: using **Lasso** to identify important features). Do the two methods agree on what features matter most? Why or why not?

Gradient Boosting Classifier (Heart Disease)	Feature Importance
age	0.050962
sex	0.021163
cp	0.286104
trestbps	0.030444
chol	0.045539
fbs	0.000000
restecg	0.004827
thalach	0.046694
exang	0.021780
oldpeak	0.116869
slope	0.042440
ca	0.170406
thal	0.162773

The most important features to the prediction were ca (number of blood vessels), thal (thallium stress result), and oldpeak (ST depression) in that order. The Linear Regression model outputted cp (carotid plaque), sex, and ca(number of blood vessels) as its most important features based on coefficients. They seem to have disagreed significantly on their most important feature selections. The Gradient Boosting Classifiers built in .feature_importances_ provides a much more robust

form of finding significant features than simply looking at model coefficients like we did with the linear regression model. This most likely explains the significant difference in their feature importance outputs.

5) Comparison to HW1
Using **only the cancer genomics dataset**:
* Select the top-K features based on gradient boosting feature importance.
* Train an HW1 model (e.g., logistic regression or Lasso-based model) using (1) the full feature set and (2) the reduced top-K feature set.
  - Did the reduced feature set improve or hurt performance?
  - Provide metrics and a short explanation of what you observed.

```
gene_18746    0.289920
gene_12983    0.202877
gene_15895    0.181088
gene_12013    0.114803
gene_18135    0.063933
gene_9176     0.055736
gene_8014     0.033764
gene_203      0.033430
gene_6160     0.006149
gene_6876     0.005088
gene_9394     0.002145
gene_5803     0.000883
gene_17904    0.000703
gene_2730     0.000655
gene_20389    0.000576
```

Top 15 features in Cancer Type Dataset based on gradient boosting feature importance

```
[Evaluation Metrics]

Accuracy Score: 0.9950248756218906
Precision Score: 0.9974999999999999
Recall Score: 0.9945945945945945
F1 Score: 0.9960024123373827
```

Model Trained on All Features

```
[Evaluation Metrics]

Accuracy Score: 0.9950248756218906
Precision Score: 0.9978494623655914
Recall Score: 0.9916666666666668
F1 Score: 0.9946635997699828
```

Model trained on top 15 features based on feature importance

The reduced feature set very slightly hurt performance. The accuracy and precision scores were almost exactly the same and the recall dropped 0.3% while the f1 score dropped 0.2%. It seems that the top 15 features capture more than 99% of the complexity of the cancer type problem and could potentially make for a more resource-efficient model that is less prone to overfitting.

6) Discussion

Discuss limitations of your models and results, including model assumptions and generalizability, potential overfitting (especially with deeper trees or large ensembles), sensitivity to hyperparameters, and data quality issues.

One significant limitation of both models is that it assumes identification is possible through several binary decisions. With healthcare data which is often very complicated and not black or white (continuous levels of certain conditions and markers) this could lead to oversimplifying decisions which actually have more complex patterns. A specific limitation of the cancer genomics model is that it assumes the distribution of genomic markers it was trained on is the same as samples in the real world. This is not necessarily always true. Additionally, Gradient Boosting Models are prone to memorizing specific gene combinations that lead to predictions rather than true underlying patterns which are more generalizable. Both models are prone to overfitting due to their large depth and large ensembles respectively (after GridSearch). Additionally, the cancer dataset is particularly prone to overfitting due to the sheer amount of data. It seems the heart disease model was much more sensitive to hyperparameters as different combinations resulted in significant differences in accuracy (as can be seen in the hyperparameter tuning table). On the other hand, the cancer genomics dataset seemed to be very insensitive to hyperparameter tuning as every combination led to almost the same accuracy, and three of the combinations led to the same highest accuracy. I don’t see any potential data quality issues as there were no missing data points and the both datasets are from reputable university studies.